



Wind turbines

Problem statement

- The costs of establishing wind farms have risen by 35% due to technological upgrades and the specific demands of site selection. Investors in this sector incur massive expenses; without a precise analysis of historical data, they risk selecting suboptimal sites or inefficient turbines, thereby reducing financial returns.



Objectives

Identify the keys that affect the decision

Identify the top counties in leading

analyze the growth of turbine capacity over the years

examine the physical evolution of hub height

compare the market share of GE Wind, Vestas, and other OEMs

Recommendations

Define keys

States

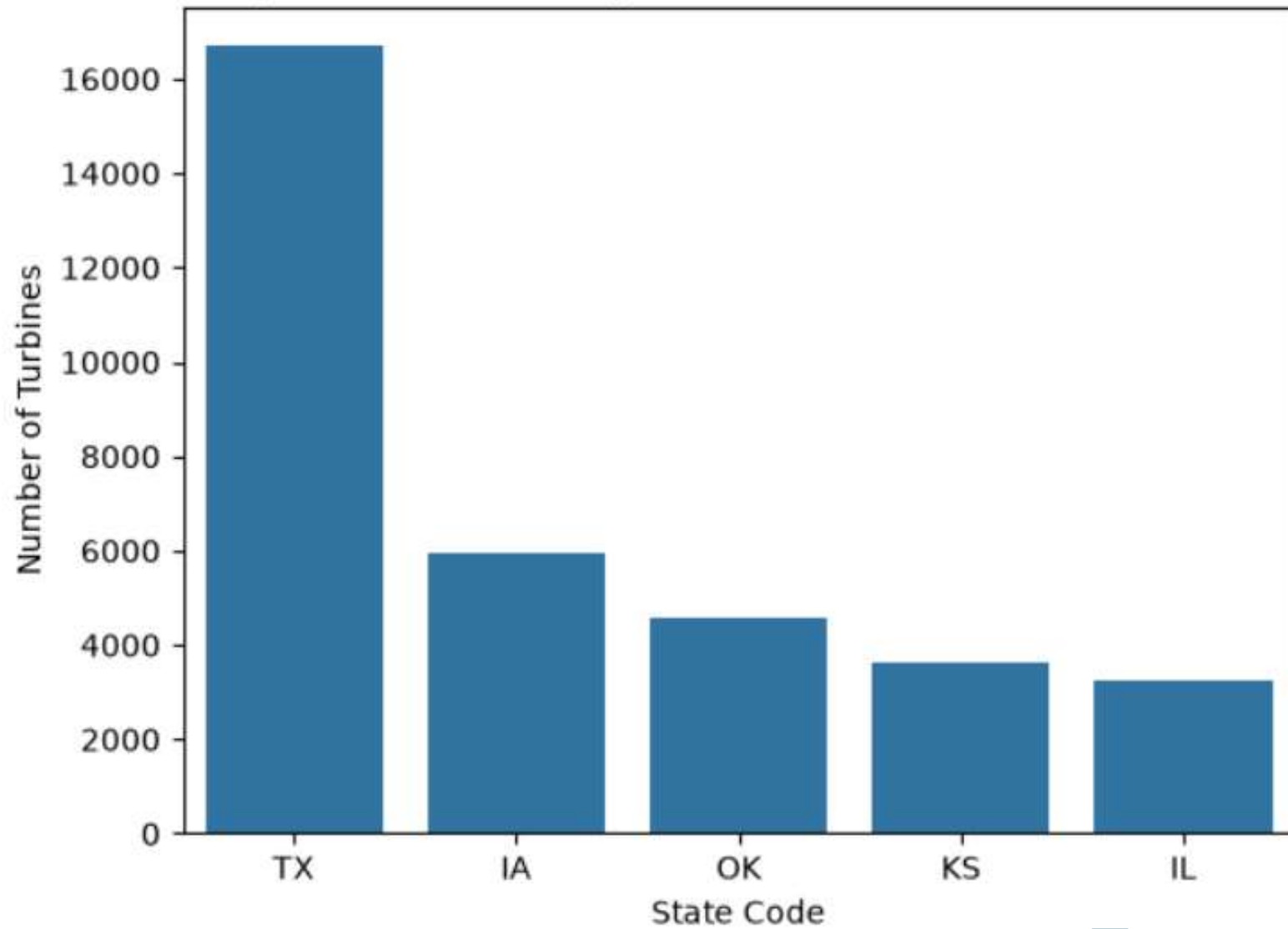
Year of production

Turbine heights

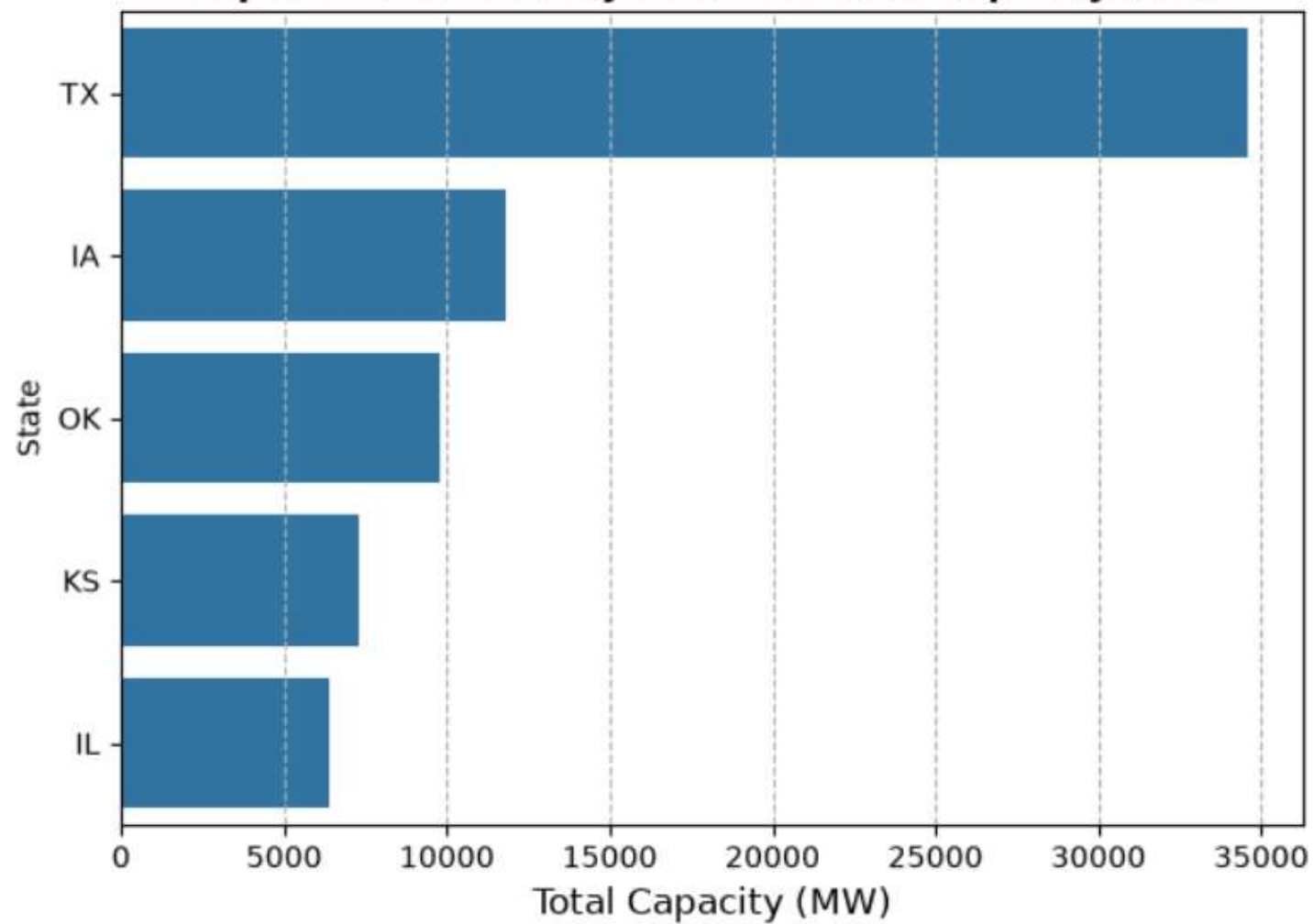
Turbine capacity

The manufacturers

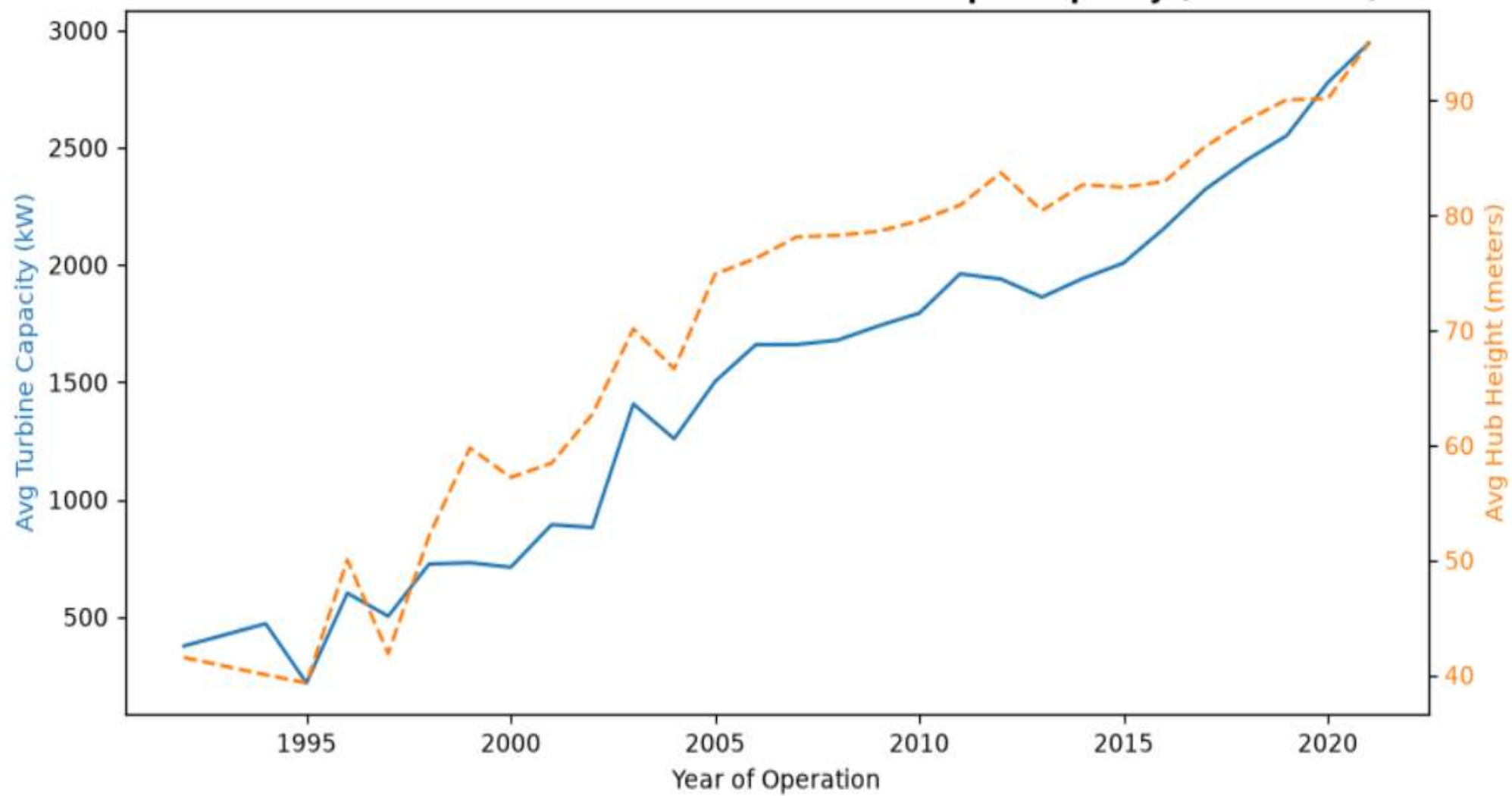
Top 5 U.S. States by Total Installed Wind Turbines



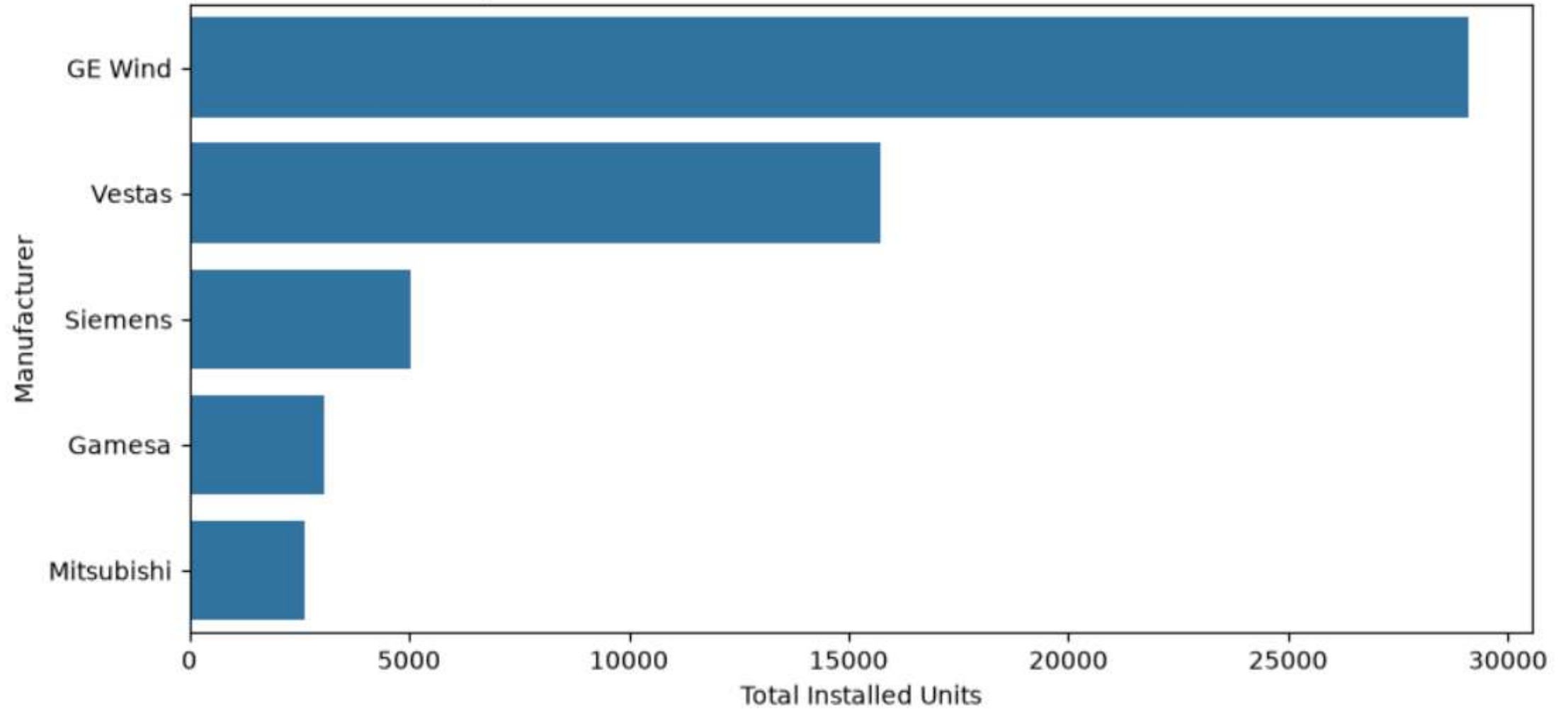
Top 10 U.S. States by Total Installed Capacity (MW)



Historical Evolution of Wind Turbine Size and Output Capacity (1990-2021)

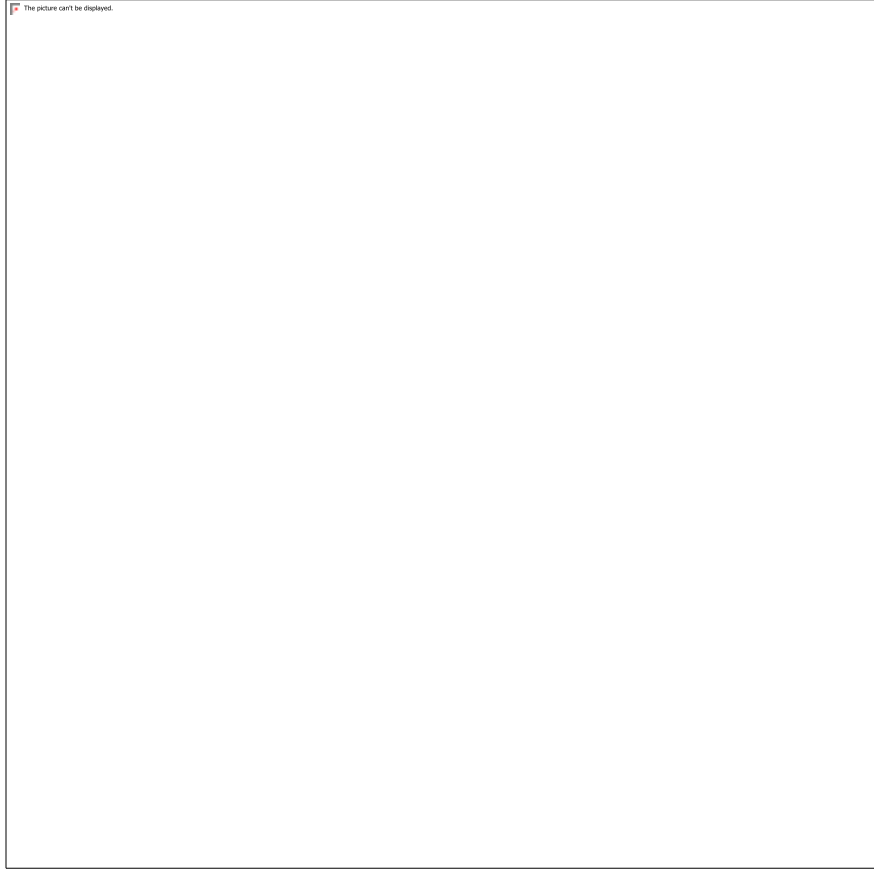


Top 5 Wind Turbine Manufacturers in the United States



Recommendation s

- **Optimize Site Selection:** Focus capital allocation on high-density hubs like Texas and the Midwest to leverage existing transmission infrastructure and streamline grid connection timelines.
- **Standardize High-Capacity Technology:** Deploy modern turbines with high capacities to maximize energy yield per square mile and offset rising capital expenditures through reduced operational footprints.
- **Mitigate OEM & Financing Risks:** Partner with Tier-1 manufacturers (GE Wind and Vestas) to secure bankable project financing and ensure long-term availability of spare parts and maintenance services



Thank You